PCB & MANUFACTURING SPECIFICATION · SKU CQA240229P1

Board Material & Fabrication Data

Bare-board specification, mechanical features and the production quality test for PiEEG-8 (board revision V2.1). Circuit design files are proprietary and are not part of this document.



FR-4	2	1.6 mm	65 × 56	0.20 mm
Base material	Copper layers	Board thickness	mm outline	Minimum track

01 BARE-BOARD FABRICATION PARAMETERS

Parameters for ordering the bare PCB, in the order used by common PCB order forms (Elecrow, JLCPCB, PCBWay).

Parameter	Specification	Note
Base material	**FR-4**, standard Tg (130–140 °C)	Glass-epoxy laminate
Layers	**2** (top and bottom copper)	
Dimensions	**65.14 × 56.20 mm**	Single board; rounded corners
Thickness	**1.6 mm**	Standard; matches Pi HAT stand-offs
Solder mask colour	**Yellow / orange**	As on production boards
Silkscreen	**White**	Pin numbers, E1–E8, Ref, Biasout, warnings
Surface finish	**Immersion gold (ENIG)** recommended	Flat pads for the fine-pitch TQFP-64; HASL lead-free acceptable
Copper weight	**1 oz** (35 µm) outer layers	
Minimum track / spacing	0.20 mm (8 mil)	Standard capability
Minimum hole	Ø0.40 mm	Vias
RoHS	Lead-free materials and process	

02 LAYER STACK-UP

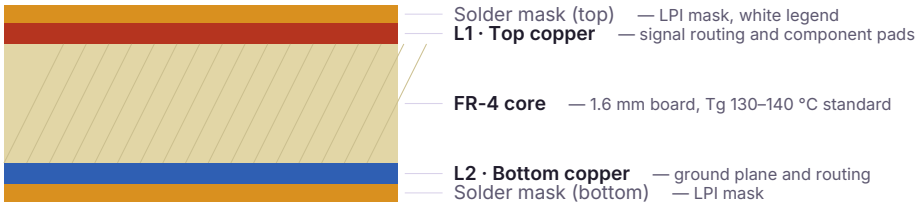


Figure 1 — 2-layer, 1.6 mm build. Layer roles are described in general terms only.

For research and education only. Not a medical device. Specifications subject to change.

03 MECHANICAL FEATURES

Feature	Size	Qty	Purpose
Mounting holes	Ø2.70 mm, non-plated area	4	M2.5 stand-offs, Raspberry Pi HAT pattern 58.0 × 48.8 mm
GPIO socket holes	Ø1.10 mm plated	40	2 × 20 stacking socket, 2.54 mm
Electrode header holes	Ø1.00 mm plated	10	1 × 10 electrode header, 2.54 mm
Jumper holes	Ø1.10 mm plated	6	JMP1–JMP3 (factory set)
Vias	Ø0.40–0.49 mm	≈ 110	Layer transitions
Board outline	65.14 × 56.20 mm	1	Profile per PE8-MD-002

04 ASSEMBLY

Technology	Mixed: surface-mount parts on the top side, through-hole connectors
Key IC	TI ADS1299 (TQFP-64), with low-noise on-board supplies
Connectors	40-pin extra-high stacking socket; 1 × 10 electrode header; 3 two-pin jumpers
Indicator	Green status LED
Soldering	Lead-free reflow for SMD, selective or hand soldering for THT
Cleanliness	No-clean flux; keep the analog input area free of flux residue
Marking	“PiEEG V2.1”, “Power supply only from battery”, “Not medical device”

05 PRODUCTION QUALITY TEST

Every assembled board is checked on a Raspberry Pi before shipping:

1 Power
Mount the shield on a Raspberry Pi 4 running Raspberry Pi OS, power from a battery. **The status LED must be on.**

2 Live signal
Run `1.8_ch_Pi4.py` from the PiEEG-8 SDK. All 8 channels must show a live trace.

3 Shorted-input check
Short **Ref** to **channel 1** and run the script again. Channel 1 must be a stable, flat line while the open channels show noise.

06 HANDLING, STORAGE AND COMPLIANCE

Topic	Requirement
ESD	Contains ESD-sensitive parts. Handle by the edges on an anti-static surface.
Storage	Dry, in the anti-static bag, 10–35 °C. Keep away from liquids and conductive gel.
Environment	Indoor use only. No conformal coating; do not operate when wet.
Classification	Development and evaluation board for research and education. Not a medical device.
Design files	Schematic, layout and BOM are proprietary to PiEEG and are not published.

07 RELATED DOCUMENTS

PE8-DS-001 Product Datasheet	**PE8-MD-002** Mechanical & Dimensional Drawing	**PE8-PIN-004** Connector Pinout & Electrode Guide
PE8-QSG-005 Quick Start & Safety Guide	**PE8-SW-006** Data Interface & Software Guide	**PE8-ENC-007** 3D-Printed Enclosure Guide